

PIPECAST Box 3 Closeout Technical Report

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This technical report chronicles the development and closeout of the flood mapping algorithm portion of the PIPECAST project. This portion of the project is referred to as Box 3. The objective of this component of the project is to take commercial SAR imagery (in this case from ICEYE), and create a binary water/nonwater map using the HYDRAFloods algorithm.

The pilot of box 3 chose to modify ICEYE SAR imagery to work in the native HYDRAFloods environment. This necessitates the upload of this imagery to Google Earth Engine. Future work could entail porting the HYDRAFloods thresholding library to a local python environment such that HYDRAFloods could be run directly on ICEYE imagery.

This work regarded the development of several .py files that modified ICEYE imagery to work with HYDRAFloods, as well as several Python notebooks that tested different parameterizations of HYDRAFloods. All of this code as well as ground truth data can be obtained from the [PIPECAST Github page](#).

Numerous tests were performed regarding pre-processing and flood mapping with the HYDRAFloods algorithm. The first test regarded the speckle filtering. Several speckle filters were tested using ground truth data including the p_median filter, gamma_map, lee_sigma, and refined_lee. The p_median filter performed best. Different parameterizations of the p_median filter, and it was observed that only the size of the pixel window used by the p_median filter contributed to the performance of the flood detection. The larger the pixel window, the better the performance. This, of course, comes with the tradeoff of coarser effective resolution.

This repository is far from complete. The geocoding module is incomplete. If this project were to be resumed, more work would be needed to accurately geocode raw SAR imagery from ICEYE. These products are provided without a geotransform.